HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed a **Python**-based agent orchestration framework on **AWS Lambda** and **API Gateway** to manage the lifecycle of AI agents, processing approximately 4,000 requests per hour.
- Engineered reusable **Python** skills and action connectors for an enterprise agent platform, enabling rapid integration of data sources and business logic, reducing new agent development time by **30%**.
- Designed and implemented **RAG** patterns using **Azure AI Search** and **LangChain** to provide agents with secure, governed access to enterprise knowledge bases, improving response accuracy by **18%**.
- Built **REST APIs** in **Java Spring Boot** for cross-service data exchange between the feature store and consumer applications, incorporating request validation and circuit-breaker patterns.
- Containerised agent platform services using **Docker** and deployed them to an **Azure Kubernetes Service (AKS)** cluster, configuring basic resource limits for predictable rollout.
- Collaborated with data engineers and business stakeholders to define requirements for AI agent capabilities, translating needs into technical specifications for prompt and tool design.
- Implemented comprehensive unit and integration tests for agent skills and orchestration logic using **pytest**, achieving over **85%** code coverage for critical components.

SKILLS

AI/ML Platform – Copilot Studio, Azure AI Foundry, Agentic AI architectures, RAG, Azure AI Search, LangChain, agent orchestration, prompt design, tool design, function apps, app services

Cloud & DevOps – Microsoft Azure, Azure Kubernetes Service (AKS), AWS Lambda, API Gateway, Docker, GitHub Actions, Azure DevOps, IaC

Backend Development – Python, TypeScript, C#, Java, Spring Boot, REST APIs, JSON, YAML, FastAPI

Data & Knowledge – Enterprise knowledge systems, data integration, data pipelines

Engineering Practices – CI/CD, secure development, testing, observability, telemetry, FinOps, Agile delivery, documentation

PROJECTS

Enterprise AI Agent Orchestration Framework (HCL Technologies)

- Developed a **Python**-based agent orchestration framework deployed on **AWS Lambda**, managing agent lifecycles and processing ~4,000 requests per hour through **API Gateway** integrations.
- Designed and implemented reusable **Python** skills and action connectors, significantly enhancing agent capabilities and streamlining integration with enterprise systems for customer onboarding.
- Integrated **RAG** patterns with **Azure AI Search** to provide agents with governed access to internal documentation, improving the accuracy of agent responses by **15%**.

Secure Enterprise Knowledge Layer for Generative AI (MSc Project)

- Built a retrieval-augmented generation (**RAG**) system using **Azure AI Search** and **LangChain** to enable AI agents to access and reason over a large corpus of enterprise documents.
- Implemented security and access controls to ensure agents only retrieve information aligned with user permissions, a critical component for regulated industries.
- Developed a **Python** API to serve embeddings and document chunks, supporting prompt engineering efforts for downstream AI model development and reducing information retrieval latency by **22%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)